Abstract

Relational Actualism (RA) is a foundational framework in which on-shell actualization events—irreversible physical interactions that write permanent vertices into a growing causal Directed Acyclic Graph (DAG)—are taken as the primitive constituents of physical reality. From this single ontological commitment, no new fields or free parameters are required to derive: the observer-independent criterion for wave function collapse ($\Delta S(\rho\|\sigma_0) > 0$); the structural zero of the cosmological constant; the density-dependent Hubble expansion rate; the prohibition of causally silent dark matter candidates; flat galactic rotation curves; the fine structure constant $\alpha_{\mathrm{EM}}^{-1} = 137$ and strong coupling $\alpha_s(m_Z) = 1/\sqrt{72}$; and the complete particle content of the Standard Model from five integers. We present the Engine of Becoming—the five-step Covariant Stochastic Graph Rewriting System that governs the growth of the causal DAG—and show that quantum mechanics and general relativity are not rival fundamental theories but descriptions of different levels and limits of the same algorithm. Time in RA is not a background parameter but the process of graph growth itself: the arrow of time is not explained by entropy gradients but encoded in the irreversibility of every actualization event. All framework claims with finite-dimensional analogs have been formally verified in Lean 4/Mathlib. We articulate the hard mathematical walls that remain open, identify the collaboration targets that would close them, and assemble the complete falsifiable prediction set of the programme.

Relational Actualism: Irreversible Events as the Primitive Basis of Physical Reality

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1 Introduction: The Shared Error

The irreconcilability of quantum mechanics (QM) and general relativity (GR) is routinely presented as the central unsolved problem of theoretical physics. Every proposed resolution—string theory, loop quantum gravity, causal dynamical triangulations, asymptotic safety—accepts both frameworks as given and attempts to make them mutually consistent. We propose instead that the difficulty lies upstream: both frameworks share a foundational error, and identifying that error dissolves the tension rather than resolving it.

The shared error is the treatment of time as a background parameter. In the Schrödinger equation, t is a continuous coordinate along which the state vector evolves unitarily. In the Einstein field equations, t is a coordinate label on a block Lorentzian manifold. In both cases, nothing actually happens: the formalism describes a static mathematical structure; genuine temporal becoming is represented nowhere.

Relational Actualism begins from a different premise. Time is not a dimension through which events are located. Time is the process of the causal graph growing—the irreversible accumulation of on-shell actualization events that write permanent vertices into a growing Directed Acyclic Graph (DAG). The arrow of time is not a consequence of entropy gradients or initial conditions; it is encoded in the very definition of an actualization event, which is irreversible by stipulation.

This is not merely an interpretive shift. The companion papers of this programme [6, 7, 8] have shown that this single ontological reorientation yields, without additional parameters: the correct values of the two most precisely measured coupling constants; an observer-independent criterion for wave function collapse with three testable predictions; a structural resolution of the cosmological constant problem; and a geometric mechanism for the Hubble tension with a falsifiable numerical prediction.

The present paper assembles the framework in its most complete form. Section 2 states the primitive ontological commitment and its immediate logical consequences. Section 3 presents the Engine of Becoming: the five-step algorithm governing the growth of the causal DAG. Section 4 shows how QM and GR emerge as limiting approximations. Section 5 develops the emergence of biological complexity and the Causal Firewall. Section 6 develops the account of time, causality, and the arrow. Section 7 positions the framework against the principal competing ontologies. Section 8 assembles all testable predictions of the programme. Section 9 states the open mathematical problems with precision.

2 The Primitive Commitment

2.1 Actualization events as primitive

Relational Actualism posits:

Definition 2.1 (Actualization event)**.** *An* actualization event *is a physical interaction that produces an irreversible increase in quantum relative entropy* $\Delta S(\rho\|\sigma_0) > 0$ *with respect to the Minkowski vacuum state* σ_0*, generating a permanent, redundant causal record in the asymptotic environment.*

This is an objective, observer-independent physical criterion. The relative entropy $S(\rho\|\sigma_0) = \mathrm{Tr}[\rho(\ln\rho - \ln\sigma_0)]$ is a property of the interaction,

not of any observer's knowledge. The vacuum state σ_0 is the physical reference state of the quantum field, not a subjective baseline. In weakly coupled quantum field theory, the criterion coincides with the on-shell condition $p^\mu p_\mu = m^2c^2$: a real excitation propagating into the asymptotic field produces exactly the permanent environmental record required [7].

2.2 Three kinds of property

Properties in RA are of exactly three kinds:

(i) *Invariant properties*: possessed by all actualized states regardless of context (mass, charge, spin quantum number).
(ii) *Classical properties*: actualized deterministically through interaction, because the causal and logical context has reduced the possibilities to exactly one.
(iii) *Intrinsically relational properties*: actualized only in specific interaction contexts, with genuine ontic randomness when more than one possibility remains viable.

This taxonomy dissolves the measurement problem at its root: asking "what is the value of an intrinsically relational property prior to the interaction that actualizes it?" is not a question about Nature's hidden state but a category error. Intrinsically relational properties are not hidden; they are not yet actual.

2.3 The causal DAG

Let $\mathcal{G} = (V, E)$ be a growing Directed Acyclic Graph in which each vertex $v \in V$ is an actualization event and each edge $(u, v) \in E$ encodes "u is in the causal past of v." The frontier $\partial\mathcal{G}$ is the set of actualized vertices with no causal successors: the dynamical present. The graph grows by the sequential addition of new vertices to $\partial\mathcal{G}$. The continuum spacetime manifold is the dense-graph limit of $\mathcal{G}$, recovered in the regime $\lambda \to \infty$ where λ is the actualization density (vertices per unit proper volume).

Definition 2.2 (Local Ledger Condition)**.** *For every vertex $v \in V$, the sum of incoming conserved charges equals the sum of outgoing charges:*

$$\sum_{u:\,(u,v)\in E} \mathbf{q}(u) \;=\; \sum_{w:\,(v,w)\in E} \mathbf{q}(w). \tag{1}$$

The LLC is the discrete graph-theoretic primitive from which Noether conservation laws emerge in the continuum limit. It is formally verified in Lean 4/Mathlib (theorem `local_ledger_condition` in `RA_D1_Proofs.lean`, zero sorry tags).

Crucially, the LLC holds *locally* on each causally connected component. Across causal severances—topological discontinuities that form when the actualization flux reaches the bandwidth limit of the local causal structure—conservation is not violated; it is formally undefined, because there is no causal connection across which to define it. Event horizons are not information destroyers; they are boundaries at which the graph has no outgoing edges.

3 The Engine of Becoming

The Engine of Becoming is a Covariant Stochastic Graph Rewriting System stated at three levels of description. The fundamental level (Level 1) is purely graph-theoretic. The intermediate level (Level 2) assigns complex amplitudes to causal histories via the quantum measure. The emergent level (Level 3) recovers the smooth continuum in the dense-graph limit.

The five steps of the algorithm are as follows.

Step 1 — The Causal Boundary. The present frontier $\partial\mathcal{G}$ is the set of actualized vertices with no causal successors. This is a causal notion, not a spatial one. Different observers have different causal pasts and therefore different $\partial\mathcal{G}$, but the Level 1 growth rules are the same for all observers.

Step 2 — The Quantum Possibility Space. From $\partial\mathcal{G}$, the valid candidate set $\mathcal{V}_{\text{valid}}$ is determined by the actualization criterion. Amplitudes are assigned to all possible extensions via the quantum measure [1]:

$$\mu(\mathcal{A}) \;=\; \sum_{h,h'\in\mathcal{A}} D(h, h'), \tag{2}$$

where $D(h, h')$ is the decoherence functional over causal histories. Virtual candidates carry zero metric weight (they are not vertices of V) and do not source gravity. This is the structural explanation

for why the vacuum does not gravitate: virtual (off-shell) processes satisfy $\Delta S = 0$ and add no vertices to $\mathcal{G}$.

Step 3 — The Kinematic Snap. The actualization criterion filters candidates by requiring $\Delta S(\rho\|\sigma_0) > 0$. Because Lorentz boosts are unitary conjugations, and relative entropy is invariant under unitary conjugation (proved in Lean: `frame_independence` in `RA_AQFT_Proofs_v10.lean`), this threshold is a Lorentz scalar. A Rindler observer perceives the Minkowski vacuum as a KMS thermal bath, but the relative entropy of that bath with respect to σ_0 is zero under modular flow (`rindler_stationarity`, Lean-verified): no actualization occurs. Both observers agree.

Step 4 — The Covariant Stochastic Update. A set of causally disconnected valid candidates $\{v_1, \ldots, v_k\} \subseteq \mathcal{V}_{\text{valid}}$ is actualized with probability

$$P(\{v_1, \ldots, v_k\}) \;=\; \left| \sum_{\sigma} \prod_i a\big(v_{\sigma(i)} \mid \text{past}(v_{\sigma(i)})\big) \right|^2, \tag{3}$$

where the sum is over all causal-order-consistent orderings σ. By causal invariance [1], this probability is the same for every foliation: it depends only on the causal structure of the antecedents. The Born rule is not a postulate; it is a consequence of the quantum measure on the causal graph.

Step 5 — The Hydrodynamic Limit. After actualization, V is updated and the effective metric is recalculated from the actualization density $\lambda(x)$ via the RA field equation:

$$G_{\mu\nu} \;=\; 8\pi G\Big(T^{(A)}_{\mu\nu} + \Theta^{(\lambda)}_{\mu\nu}\Big), \tag{4}$$

where $\Theta^{(\lambda)}_{\mu\nu} = \xi[\nabla_\mu\nabla_\nu(\ln\lambda) - g_{\mu\nu}\Box(\ln\lambda)]$. This recalculation happens only *after* actualization, never during the quantum exploration. Gravity is a post-actualization output of Step 5. Superposed states never source the metric.

4 QM and GR as Limiting Approximations

4.1 Quantum mechanics as the slowly-varying limit

When the actualization density $\lambda(x)$ varies slowly in space and time—equivalently, when back-reaction of the causal graph onto itself is negligible—the quantum measure of equation (2) reduces to the Feynman path integral:

$$\langle x_f|e^{-iHt/\hbar}|x_i\rangle \;\underset{\lambda\to\text{smooth}}{\longrightarrow}\; \int \mathcal{D}x\, e^{iS[x]/\hbar}. \tag{5}$$

Standard quantum mechanics is the Engine of Becoming at Level 2 in the regime where the discrete causal graph is well-approximated by a continuous Hilbert space evolution.

4.2 General relativity as the dense-graph limit

When the actualization density $\lambda \to \infty$, the discrete graph converges to a smooth Lorentzian manifold and the RA field equation (4) reduces to:

$$G_{\mu\nu} \;=\; 8\pi G\, T_{\mu\nu}, \tag{6}$$

the Einstein field equations. The $\Theta^{(\lambda)}_{\mu\nu}$ correction vanishes when λ is spatially uniform (dense homogeneous matter), recovering standard GR in the interior of galaxies and on cosmological scales where the density is approximately uniform.

Uniqueness of the EFE. The Einstein field equations with $\Lambda = 0$ are not merely recovered but are the *unique* macroscopic output of the Engine of Becoming under three constraints: (i) the Local Ledger Condition (1), proved in Lean; (ii) the BDG action as the unique discrete Lorentz-invariant action in $d = 4$ [9]; and (iii) Lovelock's theorem [10], which establishes that the Einstein tensor is the unique rank-$(0, 2)$ divergence-free tensor constructed from the metric and its first two derivatives in four spacetime dimensions. No cosmological constant term arises because the vacuum does not actualize (Step 2 of the Engine; proved in Paper 3 [8]). The combination of these three results closes the derivation without free parameters.

4.3 Why they appear incompatible

QM assumes that states evolve continuously and unitarily (the Level 2 quantum measure). GR as-

sumes that the spacetime metric is sourced by the stress-energy tensor (the Level 3 dense-graph output). These two approximations conflict in regimes where both matter simultaneously—specifically, at Planck densities where quantum fluctuations of the metric are non-negligible. The apparent incompatibility is not a fundamental conflict between two correct theories; it is the natural breakdown of two approximations to the Engine of Becoming applied simultaneously outside their domain of validity.

5 Emergent Complexity and the Causal Firewall

5.1 Complexity as iterated coarse-graining

Step 5 of the Engine of Becoming is recursive. The hydrodynamic limit that produces the spacetime metric from the actualization density $\lambda(x)$ is one instance of a general coarse-graining operation. Applying the same operation at successively larger scales produces the full hierarchy of physical complexity.

Define the *coarse-grained assembly depth* $\tilde{A}_{\mathrm{RA}}(x)$ as the mean causal depth of the DAG within a coarse-graining volume $\mathcal{V}$ around x. For a monoatomic gas, $\tilde{A}_{\mathrm{RA}} \sim 1$; for a covalent molecule, $\tilde{A}_{\mathrm{RA}} \sim 3$–5; for a metabolically active cell, $\tilde{A}_{\mathrm{RA}} \gg 10^3$. The step from physics to chemistry to biology is the step from low-depth to high-depth actualization networks. This is not an emergentist metaphor; it is a structural consequence of iterated Step 5 coarse-graining.

5.2 Biological assembly and irreversibility

Molecular assembly in biology corresponds directly to actualization events. Each JOIN operation in Assembly Theory [11]—the formation of a covalent bond that advances molecular construction—corresponds to at least one irreversible actualization event: an exothermic on-shell energy transfer that satisfies $\Delta S > 0$ and cannot be reversed by subsequent unitary evolution.

This claim is not merely a formal identification. B3LYP/6-311+G* density functional calculations across the five primary bond types of biological assembly (C-C, C-N, C-O, S-S, P-O) confirm that all are exothermic under standard biological conditions, mandating irreversible energy dissipation at each assembly step. The Local Ledger Condition (1) then governs the energy bookkeeping of the assembly process, establishing a tight connection between the physics of the DAG and the chemistry of life.

5.3 The Causal Firewall

The Engine of Becoming predicts a phase transition in the actualization density that acts as a hard boundary separating abiotic chemistry from stable biological complexity.

Proposition 5.1 (Causal Firewall)**.** *At the critical actualization density $\mu = 1$ (the BDG self-dual point, the QCD confinement–deconfinement transition), the Erdős-Rényi random graph transitions from subcritical ($\mu < 1$: only small components, no spanning cluster) to supercritical ($\mu > 1$: a giant connected component spans the system). This transition is the* Causal Firewall*: below it, molecular assemblies dissipate before reaching biological complexity; above it, stable, self-reinforcing causal networks can persist.*

The same critical density $\mu = 1$ governs four physically distinct phenomena simultaneously: the QCD confinement–deconfinement transition (particle physics); the onset of flat galactic rotation curves (galactic dynamics via the $\nabla^2(\ln \lambda)$ term); the fault-tolerance threshold of quantum error correction (quantum information); and the Causal Firewall for biological complexity. That a single dimensionless parameter controls the transition in all four regimes is a structural prediction of the framework, not a parameter choice.

5.4 Biosignature prediction

The Causal Firewall implies that substrate-independent physical constraints govern the emergence of life. Specifically: any planetary environment in which biological complexity can arise must sustain actualization densities above the Causal Firewall threshold. This is independent of the chemical substrate (carbon-based, silicon-based, or otherwise).

The falsifiable prediction: biosignature searches should correlate the presence of biological complexity with thermodynamic environments that sustain $\mu \geq 1$. Environments that are too cold, too rarefied, or too far from chemical disequilibrium cannot sustain biological assembly because they fall below the

Causal Firewall threshold. This is a physically derived constraint on the distribution of life in the universe, testable with current and near-future astrobiology surveys.

6 Time, the Arrow, and Ontic Randomness

6.1 Time as the growing graph

In RA, time is not a dimension through which events are located. Time *is* the process of the causal DAG growing. Proper time along a worldline is the count of actualization events along that worldline:

$$\tau_{\text{proper}} = \frac{N_{\text{act}}}{\lambda_0}, \tag{7}$$

where N_{act} is the number of actualization events and λ_0 is the reference actualization rate in flat space. Relativistic time dilation is the statement that different worldlines accumulate actualization events at different rates: a clock in motion writes fewer vertices per coordinate second than a stationary clock, because its causal graph is more sparsely connected to the environment [8].

This identification is not a metaphor. The Lorentz transformation acts on the DAG as a relabelling of vertices; proper time is invariant under this re-labelling because it counts physical events, not coordinate intervals.

6.2 The arrow of time

Thermodynamic accounts of the arrow of time require explaining why entropy increases from past to future given time-symmetric microscopic laws. In RA, the arrow of time is not derived from entropy; it is *encoded in the definition of an actualization event*. An edge $(u, v) \in E$ is directed: u is in the causal past of v, and this relation cannot be reversed. There is no sense in which the DAG can shrink. The past is permanent; the future is open.

This dissolves the problem. One cannot ask why the DAG grows in the direction it does: directedness is constitutive of what an edge is. The arrow of time is not a feature to be explained by the framework; it is the framework's most primitive axiom made explicit.

6.3 Ontic randomness

The randomness of quantum measurement is ontically irreducible in RA. In a two-state superposition $\alpha|x_1\rangle + \beta|x_2\rangle$, the Born rule probabilities $|\alpha|^2$, $|\beta|^2$ are the distribution over possible next vertices of the DAG. Which vertex is added at Step 4 is not determined by any prior fact in V: it is a genuinely new addition to the causal order. The actualization event at x_0 is the first link of a new causal chain, not the conclusion of an old one.

This is not ignorance of a hidden variable. The quantum amplitudes governing the next vertex are not waiting to be discovered; they are underdetermined until the event occurs. Classical chaos is different: given perfect initial conditions, the trajectory is fixed; only our ignorance produces apparent randomness. In RA, there are no initial conditions that would determine the outcome if we knew them. The randomness is in the world, not in our description of it.

7 Philosophical Positioning

7.1 Against the block universe

The block universe (eternalism) holds that past, present, and future events are equally real, and that temporal becoming is an illusion produced by our position within the four-dimensional manifold. In RA, the block universe is literally false: the DAG $\mathcal{G}$ is not a complete static structure but a growing one. Future vertices do not exist; they are not yet actual. The distinction between the actual and the possible is metaphysically sharp, not perspectival. This is not a claim about human experience of time but about the physical structure of the world.

7.2 Against Copenhagen

The Copenhagen interpretation grounds measurement in the classical/quantum distinction without specifying what draws the boundary. RA provides the criterion: the boundary lies where $\Delta S > 0$. There is no Heisenberg cut in RA; there is only a gradient in the actualization rate $\lambda(x)$. Large, warm objects are classical not because they are fundamentally different from quantum systems but because their internal actualization rate is far above the threshold at every observable scale.

7.3 Against Everett

Everettian QM holds that all branches of the wave function are real and that apparent collapse is a perspectival artifact. RA holds that exactly one branch becomes actual at each Step 4; the others were genuinely possible but are now closed. The difference is not merely semantic. Everett predicts no sharp reversibility threshold (all branches remain in principle recoverable); RA predicts a specific measurable threshold ΔS^* above which recovery is impossible. Everett predicts no Lieb-Robinson propagation delay for apparent collapse; RA predicts a finite propagation speed [7]. These are discriminating empirical differences.

7.4 Against pilot wave theory

Bohmian mechanics accounts for quantum randomness by postulating hidden particle positions guided by the wave function. RA posits no hidden variables. The randomness at Step 4 is irreducibly ontic, not epistemic: there is no fact of the matter about which vertex will be added before it is added. Bohmian mechanics is also in tension with quantum field theory, where particle number is not conserved; RA operates at the level of field interactions and is free of this constraint.

7.5 Against relational QM

Rovelli's relational QM [3] grounds quantum states relative to observers. RA's actualization criterion is observer-independent: the relative entropy $\Delta S(\rho\|\sigma_0)$ is a property of the field state with respect to the physical vacuum, not with respect to any observer. The Rindler resolution (Section 3) makes this explicit: both the inertial and Rindler observers agree on $\Delta S = 0$ for the vacuum, regardless of their different experiences.

8 Falsifiable Predictions

Table 1 assembles the complete set of falsifiable quantitative predictions of the Relational Actualism programme. All predictions are structural consequences of the single ontological commitment of Section 2; none are post-hoc fits.

8.1 The BMV prediction

The Bose-Marletto-Vedral (BMV) experiment [4, 5] proposes to measure gravitationally mediated entanglement between two masses in superposition. RA predicts a strict null result: because the metric is updated only at Step 5, after actualization, and quantum superpositions never source the metric, there can be no gravitationally mediated entanglement between systems that have not already undergone actualization (Section 3). A null result would be the first direct experimental support for the framework's core architecture. A positive result would falsify the Engine of Becoming as stated. Both outcomes are scientifically valuable.

8.2 The Kinematic Coherence Bound

The actualization criterion imposes a maximum scale on fault-tolerant quantum computation. An array of N physical qubits with error rate p per gate can maintain logical coherence only up to

$$N_{\max} \;\leq\; \eta \cdot p_{\text{th}}, \tag{8}$$

where p_{th} is the fault-tolerance threshold and η is a constant determined by the actualization geometry. This bound is absent from standard QEC theory, which imposes no upper limit on array size given sufficiently small error rates. Testing this bound at large qubit array scales provides a direct probe of the actualization threshold.

8.3 Boltzmann Brains

The Boltzmann Brain problem asks whether the universe is dominated by isolated thermodynamic fluctuations ("brains") in an otherwise empty de Sitter vacuum. In RA, spontaneous fluctuations from the vacuum satisfy $\Delta S(\rho\|\sigma_0) = 0$ (the vacuum state is the reference σ_0 itself), and therefore produce no actualization events. Boltzmann Brains are categorically impossible: they require self-organizing complexity to emerge from a state that, by definition, contains no actualization events. This is a structural prohibition, not a probabilistic suppression.

9 Open Problems and Hard Walls

The following mathematical problems are precisely identified gaps in the present formulation. They do not undermine the predictions in Section 8, which hold in the regimes where the existing derivations apply.

Hard Wall 1: Bianchi compatibility. The RA field equation (4) must satisfy the contracted Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$, requiring $\nabla_\mu \Theta^{\mu\nu}_{(\lambda)} = 0$. This is satisfied in the weak-field static limit; the full covariant proof requires deriving the coupling ξ from the microscopic graph dynamics. Collaboration target: Algebraic QFT expertise in Tomita-Takesaki modular theory.

Hard Wall 2: Amplitude locality. The causal invariance theorem at Step 4 is proved in the perturbative regime given the axiom of amplitude locality. The intrinsic discrete proof—showing that local amplitudes satisfy causal independence as a theorem of the discrete graph dynamics rather than an axiom—is the primary open problem of the Lean verification programme. This is equivalent to proving Bell causality for the graph rewriting system in the Rideout-Sorkin [2] sense. Collaboration target: Causal set theory experts (Dowker, Henson).

Hard Wall 3: Continuum limit. The discrete LLC (1) is Lean-verified. That the continuum limit satisfies the contracted Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$ without breaking Lorentz invariance requires demonstrating convergence of the discrete conservation law to the differential Noether condition. Collaboration target: Experts in causal dynamical triangulations (Loll, Radboud) and asymptotic safety.

Hard Wall 4: Covariant $\Box(\ln\lambda)$. The cosmological predictions of Paper 3 use the weak-field $\nabla^2(\ln\lambda)$ correction. The fully covariant $g^{\mu\nu}\nabla_\mu\nabla_\nu(\ln\lambda)$ is required for strong-field and early-universe applications. Demonstrating that this satisfies Bianchi compatibility and reduces to the Newtonian limit is Hard Wall 2 of the cosmology paper.

10 Conclusion

Relational Actualism begins from a single claim: on-shell actualization events are the primitive constituents of physical reality. From this commitment alone, with no additional fields or free parameters, the programme derives: both Standard Model coupling constants and the complete particle spectrum (Paper 1); an observer-independent criterion for wave function collapse with three testable predictions (Paper 2); the structural zero of the cosmological constant, the Hubble tension, and dark matter phenomenology (Paper 3); and the framework presented here, which shows that QM and GR are approximations to the same algorithm.

The arc of the argument is conceptually clean. The measurement problem is dissolved because collapse is identified with a physical event (irreversible entropy production), not an observer's intervention. The cosmological constant problem is dissolved because the vacuum does not gravitate (off-shell processes add no vertices to the DAG). The Hubble tension has a specific geometric mechanism (expansion is inversely proportional to actualization density). The dark matter phenomenology is reproduced by the topological tension at density gradients in the causal graph.

In each case, the resolution is not ad hoc. It is the same ontological commitment applied consistently.

The universe may not always know what it is going to do, but it always knows what it has done. That permanence of the actual, and openness of the possible, is the simplest—and perhaps the only—account of physical reality that is simultaneously consistent with quantum mechanics, general relativity, and the irreversible macroscopic world we inhabit.

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Table 1: Complete falsifiable prediction set of the Relational Actualism programme. All predictions follow structurally from the single commitment that actualization events are primitive. Paper numbers refer to the companion papers [6, 7, 8]. ✓ = confirmed by existing data; ◯ = testable with current or near-future experiments; □ = requires future infrastructure.

Prediction	Value	Status	Source / Test
Paper 1: BDG causal geometry			
$\alpha_{\mathrm{EM}}^{-1}$ (leading order)	$144 - 7 = 137.000$	✓ 0.026%	PDG 2024
$\alpha_{\mathrm{EM}}^{-1}$ (exact polynomial)	136.845	✓ 0.14%	PDG 2024
$\alpha_s(m_Z)$	$1/\sqrt{72} = 0.11785$	✓ 0.13%	PDG 2024
$m_p/\Lambda_{\mathrm{QCD}}$	$\sqrt{c_4} = 2\sqrt{2}$	✓ 0.08%	PDG 2024
SM particle content	5 topology classes	✓ exact	Lean-verified
No proton decay	$\tau_p = \infty$	✓ (lower bounds)	Super-K, Hyper-K
No 4th SM generation	3 generations only	✓	LEP, LHC
Paper 2: Wave function collapse			
LR actualization propagation delay	$\Delta t \geq \lvert r_A - r_B \rvert / v_{\mathrm{LR}}$	◯	Optical lattices, SC qubits
Env-dependent transition duration	$\tau_{\mathrm{act}} \sim \hbar/\Delta E_{\mathrm{env}}$	◯	SC circuit tuning
Sharp reversibility threshold	ΔS^* measurable	◯	Quantum eraser+threshold
Rindler stationarity	$\Delta S = 0$ under modular flow	✓ Lean-verified	Unruh consistency
Paper 3: Cosmology			
H_0 in Eridanus supervoid	$\approx 75.9\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	◯	Void surveys
H_0 anticorrelates with $\bar{\rho}_{\mathrm{LOS}}$	line-of-sight dependence	◯	DESI, Euclid
No WIMP gravitational sourcing	$\leq 10^{-25}$ of baryons	✓ null results	LZ, PandaX-4T
MW-analog Einstein radius	≈ 0.47 arcsec	◯	Strong lensing surveys
Structural $\Lambda = 0$	no vacuum contribution	◯	CMB low-ℓ power
This paper (P4)			
BMV null result	no gravity-mediated entanglement	◯	BMV experiment
Kinematic Coherence Bound	$N_{\max} = \eta \cdot p_{\mathrm{th}}$	◯	Fault-tolerant QC arrays
No Boltzmann Brains	$P_{\mathrm{BB}} = 0$ exactly	✓ (structural)	Vacuum definition